

YELP RESTAURANT BUSINESS PERFORMANCE ANALYSIS REPORT

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1. Introduction

The restaurant industry is highly competitive, and customer engagement plays a major role in determining business success. Online platforms such as Yelp provide valuable data regarding customer reviews, ratings, check-ins, and interactions.

This project focuses on analyzing Yelp restaurant data to identify the major factors affecting business performance. Using SQL and Python-based Exploratory Data Analysis (EDA), this report extracts meaningful business insights from customer behavior patterns.

The objective is to understand how customer engagement metrics influence restaurant performance and success.

2. Business Problem

Restaurants often face challenges in understanding the factors that drive customer satisfaction and business growth.

The major business questions addressed in this project are:

- What factors influence restaurant success?
- How do customer reviews affect business performance?
- What is the relationship between customer engagement and ratings?
- During which hours is customer activity highest?
- How do elite users contribute to engagement?

This analysis helps businesses optimize operations, improve customer experience, and increase profitability.

3. Dataset Overview

The dataset used in this analysis is the Yelp dataset containing multiple tables.

Main Tables Used

Business Table

Contains restaurant/business information:

- Business ID
- Name
- City
- State
- Stars
- Review Count
- Categories

Review Table

Contains customer review details:

- Review ID
- Business ID
- Stars
- Useful Votes
- Funny Votes
- Cool Votes
- Review Date

Tip Table

Contains customer tips.

Check-In Table

Contains customer visit/check-in information.

User Table

Contains user details including elite and non-elite users.

4. Tools and Technologies Used

The following tools were used for this project:

- MySQL
 - Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Statsmodels
 - Jupyter Notebook
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5. Data Cleaning and Preprocessing

The following preprocessing steps were performed:

- Connected MySQL database with Python

- Loaded data into Pandas DataFrames
- Handled missing values
- Converted date columns to proper datetime format
- Removed unnecessary columns
- Aggregated engagement metrics
- Created business success score

These steps ensured accurate analysis and improved data quality.

6. Exploratory Data Analysis (EDA)

6.1 Rating Trend Analysis

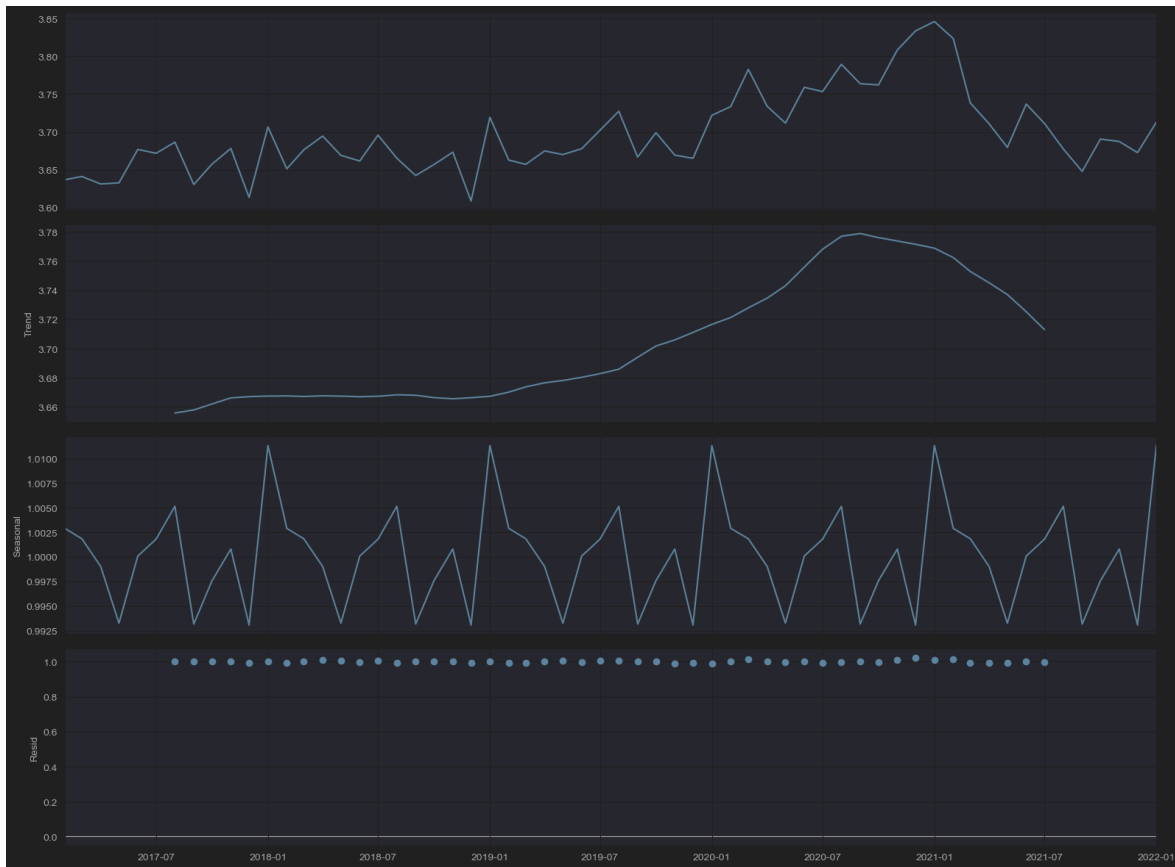
Seasonal decomposition was performed to analyze rating trends over time.

Observations:

- Ratings remained relatively stable.
- Most restaurants maintained ratings between 3.5 and 4 stars.
- Small fluctuations were observed over time.

Insight:

Restaurant ratings remain consistent, indicating relatively stable customer satisfaction.



6.2 Tip Engagement Analysis

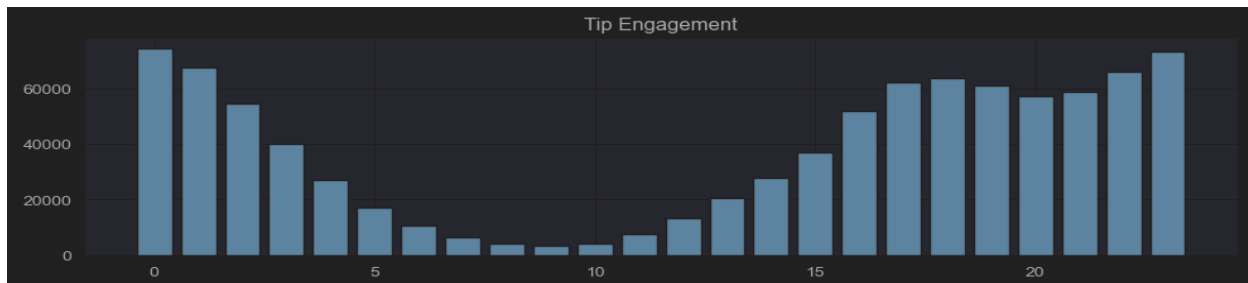
Tip engagement was analyzed by hourly activity.

Observations:

- Highest tip activity observed during lunch and dinner hours.
- Peak engagement occurs in evening time.

Insight:

Customer interaction through tips increases significantly during peak dining hours.



6.3 Review Engagement Analysis

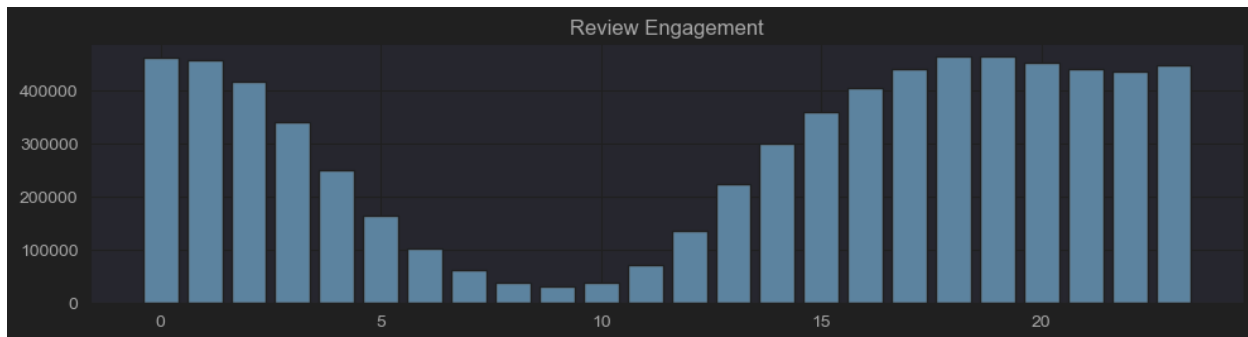
Review activity was analyzed across different hours.

Observations:

- Review activity is highest during evening hours.
- Customers are more likely to leave reviews after dining.

Insight:

Evening hours generate the highest review engagement.



6.4 Check-In Engagement Analysis

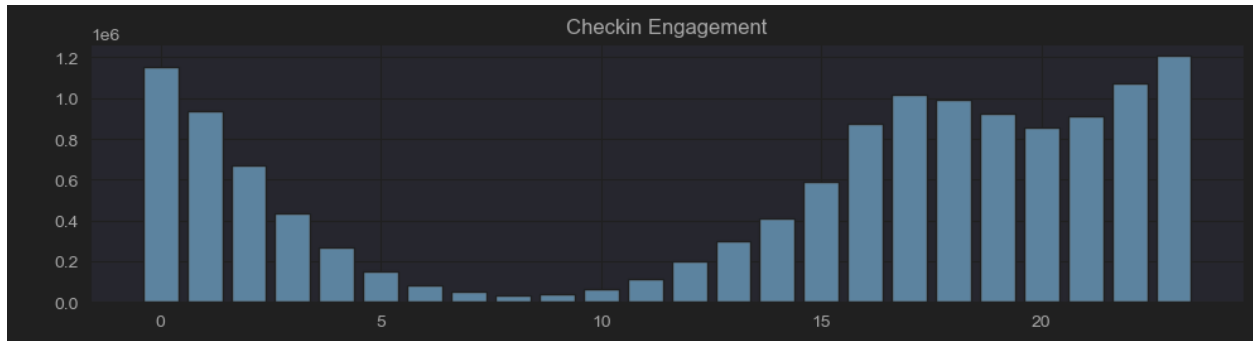
Check-in patterns were analyzed.

Observations:

- Check-ins peak during lunch and dinner timings.
- Morning hours show lower activity.

Insight:

Business traffic is highest during meal hours.



6.5 Elite vs Non-Elite User Analysis

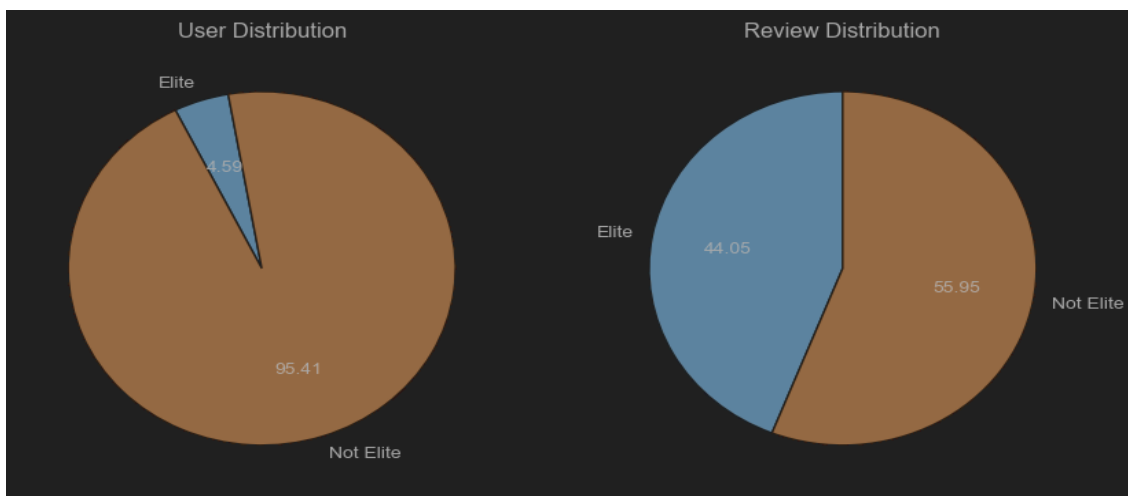
User contribution was compared between elite and non-elite users.

Observations:

- Elite users represent a small percentage of total users.
- Despite smaller numbers, elite users contribute significantly to engagement.

Insight:

Elite users are highly valuable for businesses because they contribute disproportionately high engagement.



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6.6 Correlation Analysis

A correlation heatmap was created to understand relationships between variables.

Key variables:

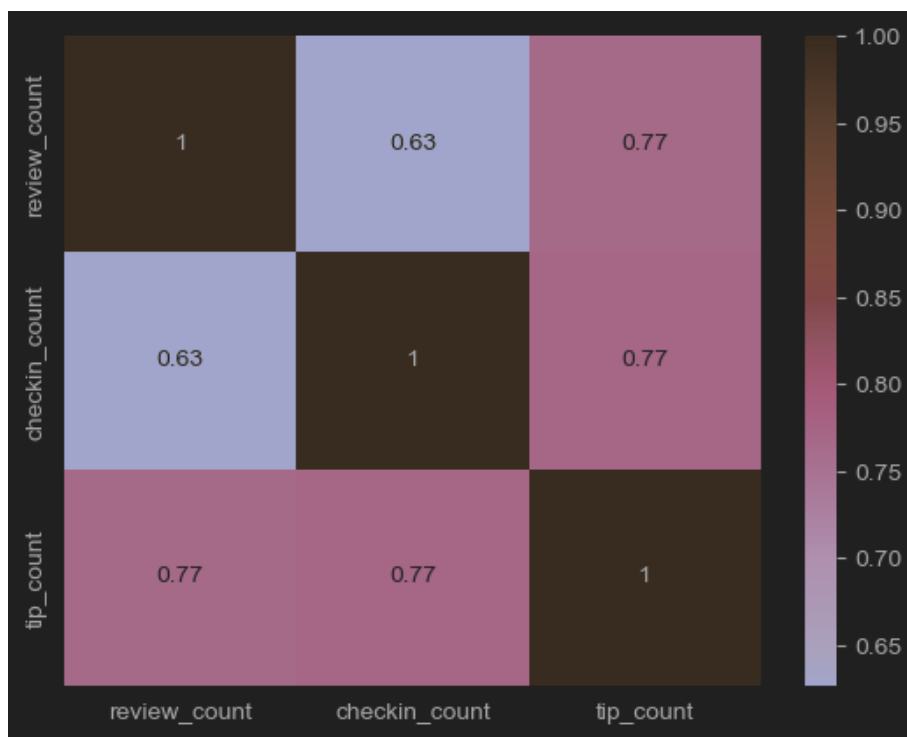
- Review Count
- Useful Votes
- Funny Votes
- Cool Votes
- Success Score

Observations:

- Review count strongly correlates with success score.
- Useful votes also show strong positive correlation.
- Cool votes positively impact business performance.

Insight:

Higher engagement generally leads to higher business success.



7. Key Insights

Major findings from this analysis:

- Peak customer activity occurs during lunch and dinner hours.

- Ratings remain relatively stable over time.
 - Reviews strongly influence business performance.
 - Elite users contribute valuable engagement.
 - Useful and cool votes positively impact business success.
 - Higher customer engagement improves restaurant performance.
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8. Business Recommendations

Based on analysis, the following recommendations are suggested:

Recommendation 1

Focus promotions during peak engagement hours:

- Lunch hours
 - Dinner hours
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Recommendation 2

Encourage customers to leave reviews.

More reviews improve visibility and business success.

Recommendation 3

Target elite Yelp users through promotions and loyalty programs.

Elite users contribute high-value engagement.

Recommendation 4

Improve customer experience to increase:

- Positive reviews
 - Useful votes
 - Customer retention
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9. Conclusion

This project successfully analyzed Yelp restaurant business performance using SQL and Python-based Exploratory Data Analysis.

The analysis revealed that restaurant success is strongly influenced by:

- Customer reviews
- Check-in activity
- User engagement
- Elite customer contribution

Businesses with strong customer engagement and better review performance are more likely to achieve higher ratings and long-term success.

This project demonstrates how data analytics can help businesses make informed decisions and improve operational performance.